CSDA5110
Spring 2024
Logistic Regression

1. Introduction

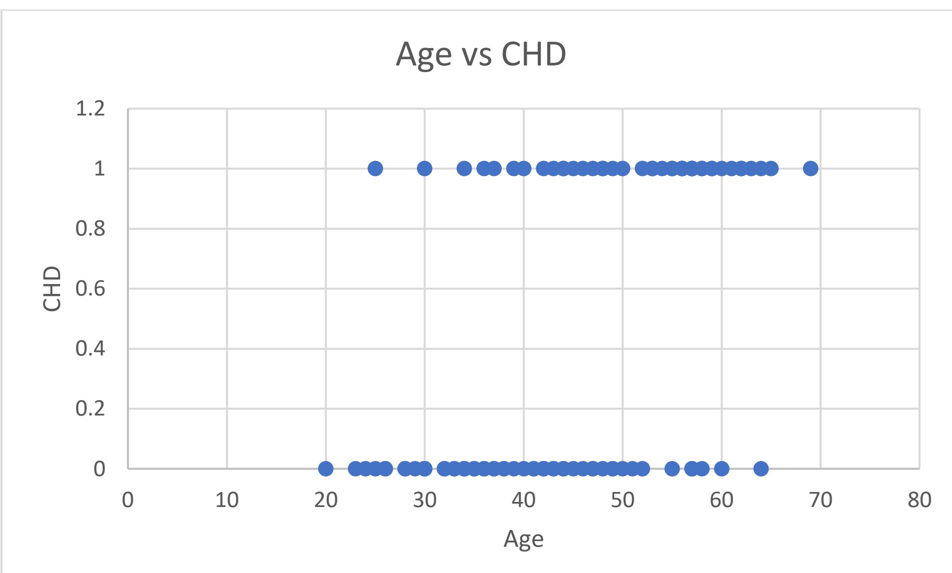
Objective:
Same as regression model – model relationship between response variable and set of predictor variables

Difference:

- Outcome variable in Logistic Regression is bivariate or dichotomous (Y_i =1 if event occurs or Y_i =0 if event does not occur)
- Choice of parametric model
- Assumptions

Follow the same general principles used in linear regression.

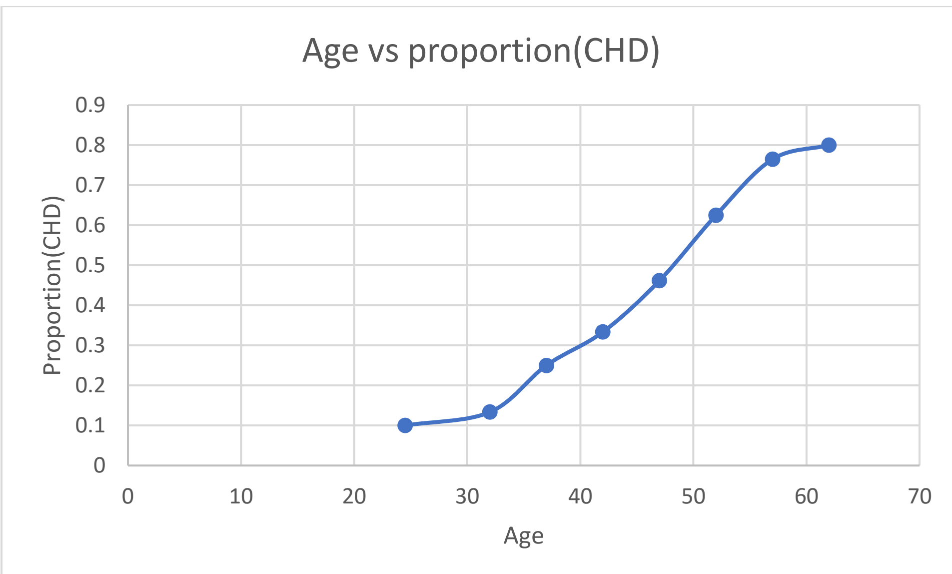
Example: presence or absence of evidence of significant coronary heart disease (CHD) and age for 100 subjects



- Two parallel lines
- Variability in CHD at all ages is large
- Makes it difficult to describe the functional relationship between age and CHD

Frequency Distribution

Age Group	midvalue	frequency	CHD_present	CHD_absent	Proportion_CHD
20-29	24.5	10	1	9	0.10
30-34	32	15	2	13	0.13
35-39	37	12	3	9	0.25
40-44	42	15	5	10	0.33
45-49	47	13	6	7	0.46
50-54	52	8	5	3	0.63
55-59	57	17	13	4	0.76
60-69	62	10	8	2	0.80
		100	43	57	0.43



- As age increases, the proportion of individuals with evidence of CHD increases
- The plot gives insight into relationship between the two variables
- Describe the functional form of the relationship
    - In regression analysis, mean value of the response variable given the value of the independent variable is the main quantity $\{E(Y|X=x) = \beta_0 + \beta_1 x\}$.
    - In dichotomous date, $E(Y|X=x)$ must be greater than or equal to 0 and less than equal to 1.
    - Plot shows that change in $E(Y|x)$ per unit change in x becomes progressively smaller as the conditional mean gets closer to 0 or 1.
    - The curve is said to be *S*-shaped.

Logistic Distribution

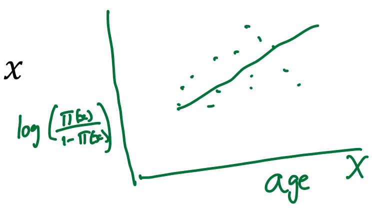
- Extremely flexible and easily used function
- Lends itself to meaningful interpretation

Logistic Regression Model

$$E(Y|x) = \pi(x) = \frac{e^{\beta_0+\beta_1 x}}{1+e^{\beta_0+\beta_1 x}}$$

Link Function : The log of the odds that an event occurs, otherwise known as the "logit"

$$\text{logit}(\pi(x)) = \log\left(\frac{\pi(x)}{1-\pi(x)}\right) = \beta_0 + \beta_1 x$$



- $\text{logit}(\pi(x))$ is linear in its parameters
- It has many of the desirable properties of linear regression
- The plot of $\text{logit}(\pi(x))$ against x will produce a straight line

- In regression model, $y = E(Y|x) + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$

- For dichotomous response variable, $y = \pi(x) + \epsilon$.
  - ϵ may assume value $1-\pi(x)$ with probability $\pi(x)$ if y=1
  - ϵ may assume value $-\pi(x)$ with probability $1-\pi(x)$ if y=0
  - Hence ϵ has a distribution with mean 0 and variance $=\pi(x) * \{1-\pi(x)\}$
  - Conditional distribution of the outcome variable follows a Binomial Distribution with probability given by $\pi(x)$

Summary

- The conditional mean of the regression equation must be formulated to be bounded between 0 and 1. $\pi(x)$ satisfies this constraint
- The binomial distribution describes the distribution of the errors and will be the statistical distribution upon which the analysis is based (unlike normal distribution in linear regression)
- The principles that guide an analysis using linear regression will also guide analysis of logistic regression

2. <u>Fitting the Logistic Regression</u>

- To fit the logistic regression to a given data requires estimation of the values of β_0 & β_1, the unknown parameters
- Ordinary Least Squares (OLS) chose the values of β_0 & β_1 which minimize the sum of squared deviations of the observed values of y from the predicted values based upon the model
- For logistic regression, the likelihood equations are:
    - $\sum_i^n [y_i - \pi(x_i)] = 0$ (1)
    - $\sum_i^n x_i [y_i - \pi(x_i)] = 0$ (2)
- The expressions in (1) & (2) are non-linear in β_0 & β_1 and therefore, require special methods to obtain solution.
- Will use software to obtain solutions
- $\hat{\pi}(x)$ is the maximum likelihood estimate of $\pi(x)$
    - This quantity provides an estimate of the conditional probability that Y is equal to 1 given that X = x
    - It represents the fitted or predicted value for the logistic regression model
    - $\sum y_i = \sum \hat{\pi}(x)$
    - $\widehat{\text{logit}(\pi(x))} = \log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = b_0 + b_1 x$

Example 1

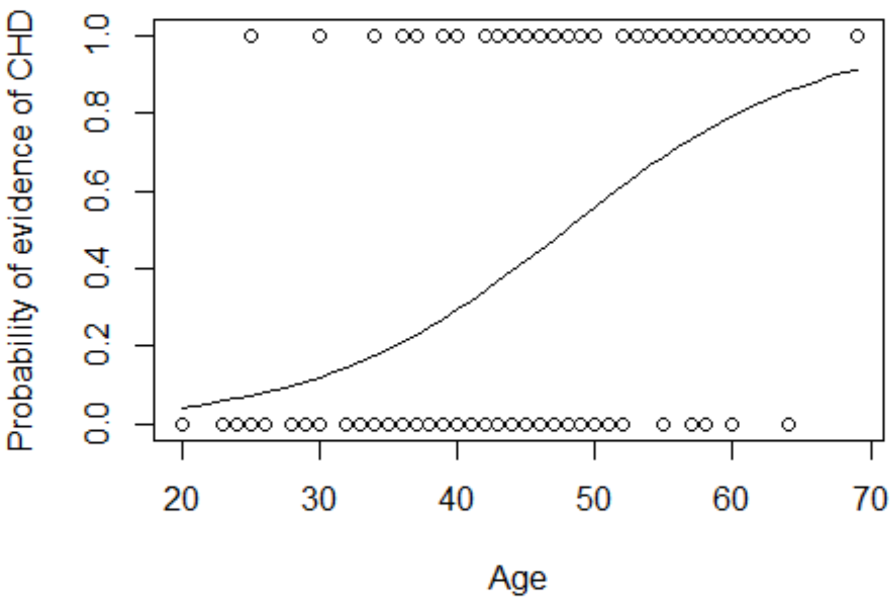
```r
output_LR1 <- glm(CHD~Age,family=binomial, chddat)
summary(output_LR1)
```

```
##
## Call:
## glm(formula = CHD ~ Age, family = binomial, data = chddat)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.30945    1.13365  -4.683 2.82e-06 ***
## Age          0.11092    0.02406   4.610 4.02e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 136.66  on 99  degrees of freedom
## Residual deviance: 107.35  on 98  degrees of freedom
## AIC: 111.35
```

$\hat{\pi}(x) = \frac{e^{b_0+b_1x}}{1+e^{b_0+b_1x}}$

PLot

```r
attach(chddat)
x <- seq(20,69,1)
y <- 1/(1+exp(-1*(output_LR1$coeff[1] + output_LR1$coeff[2]*x)))
plot(Age, CHD,ylab="Probability of evidence of CHD",xlab="Age")
lines(x,y)
```



Fitted Values

```r
pihat <- output_LR1$fitted.values
odds_ratio <- output_LR1$fitted.values/(1-output_LR1$fitted.values)
cbind(Age,round(pihat,3),round(odds_ratio,3))
```

```
##     Age
## 1    20 0.043  0.045
## 2    23 0.060  0.063
## 3    24 0.066  0.071
## 4    25 0.073  0.079
## 5    25 0.073  0.079
## 6    26 0.081  0.088
## 7    26 0.081  0.088
## 8    28 0.099  0.110
## 9    28 0.099  0.110
## 10   29 0.110  0.123
## 11   30 0.121  0.138
## 12   30 0.121  0.138
## 13   30 0.121  0.138
```

$$\text{logit}\left[\hat{\pi}(x)\right] = \log\left(\frac{\hat{\pi}(x)}{1-\hat{\pi}(x)}\right) = b_0+b_1x$$

$$\text{logit}\left[\hat{\pi}(x+1)\right] - \text{logit}\left[\hat{\pi}(x)\right]$$

$$= b_1$$

$$\ln\left(\frac{\text{odd }2}{\text{odd }1}\right) = b_1$$

$$\text{odds ratio} = e^{b_1}$$

Interpretation of b_1

- log(odds)
- odds ratio: e^{b_1} = 1.12

➢ odds increase by 12%with each additional year of age

Test for $\boldsymbol{\beta}$: Wald's test statistic $Z = \frac{b_1}{\widehat{SE}(b_1)}$

Compare Z to standard normal distribution

Deviance:

In logistic regression the concept of the residual sum of squares is replaced by a concept known as the deviance.

The deviance associated with a given logistic regression model is based on comparing the maximized log-likelihood of the fitted model with the maximized log-likelihood under the so-called saturated model that has a parameter for each observation. In fact, the deviance is given by twice the difference between these maximized log-likelihoods.

$$D^2 = -2ln\left[\frac{(likelihood\ of\ the\ fitted\ model)}{(likelihood\ of\ the\ saturated\ model)}\right]$$

$D^2 = -2\sum_{i=1}^{n}\left[y_i * ln\left(\frac{\hat{\pi}_i}{y_i}\right) + (1-y_i) * ln\left(\frac{1-\hat{\pi}_i}{1-y_i}\right)\right]$, where $\hat{\pi}_i = \hat{\pi}(x)$

Likelihood of saturated model is 1

H_0 : logistic regression model is appropriate
H_1 : logistic model is inappropriate so a saturated model is needed

Under the null hypothesis, the deviance D^2 is approximately distributed as $\chi^2_{(n-p-1)}$, where n = the number of binomial samples, p = number of predictors in the model

```
## Residual deviance: 107.35  on 98  degrees of freedom
```

```
pchisq(output_LR1$deviance,output_LR1$df.residual,lower=FALSE)
```

```
## [1] 0.243462
```

we are unable to reject H_0. In other words, the deviance goodness-of-fit test finds that the logistic regression model is an adequate fit overall for the CHD data.

Difference of Deviance to compare models Partial Sum of Squares

For the purpose of assessing the significance of an independent variable, we compare value of D with and without the independent variable in the equation.

$$G^2 = D(model\ without\ the\ variable) - D(model\ with\ the\ variable)$$

$$G^2 = -2ln\left[\frac{(likelihood\ without\ the\ variable)}{(likelihood\ with\ the\ variable)}\right]$$

$$G^2 = -2ln\left[\frac{\left(\frac{n_1}{n}\right)^{n_1} * \left(\frac{n_o}{n}\right)^{n_0}}{\prod_{i=1}^{n} \hat{\pi}_i^{y_i}(1-\hat{\pi}_i)^{(1-y_i)}}\right]$$

Under the null hypothesis, $H_0 : \beta_1 = 0$, the deviance G^2 is approximately distributed as $\chi^2_{(1)}$, where n should be large.

```
##     Null deviance: 136.66  on 99  degrees of freedom
## Residual deviance: 107.35  on 98  degrees of freedom
```

$H_0: \beta_1 = 0$

##Value of the difference in devinace and associated p-value

```
output_LR1$null.deviance-output_LR1$deviance

## [1] 29.30989

pchisq(output_LR1$null.deviance-output_LR1$deviance,1,lower=FALSE)

## [1] 6.168008e-08
```

$< \alpha = .05$

reject Ho

R^2 for Logistic Regression

$$R^2_{dev} = 1 - \frac{D^2_{H_1}}{D^2_{H_0}} = \frac{107.35}{136.66} = .2145$$

Interpretation

$\hat{\beta}_1 = .111$ Change in log-odds of CHD per one year increase in age

odds ratio

$$e^{\hat{\beta}_1} = e^{.111} = 1.117$$

odds of CHD increases by 11.7%
Per year increase in age